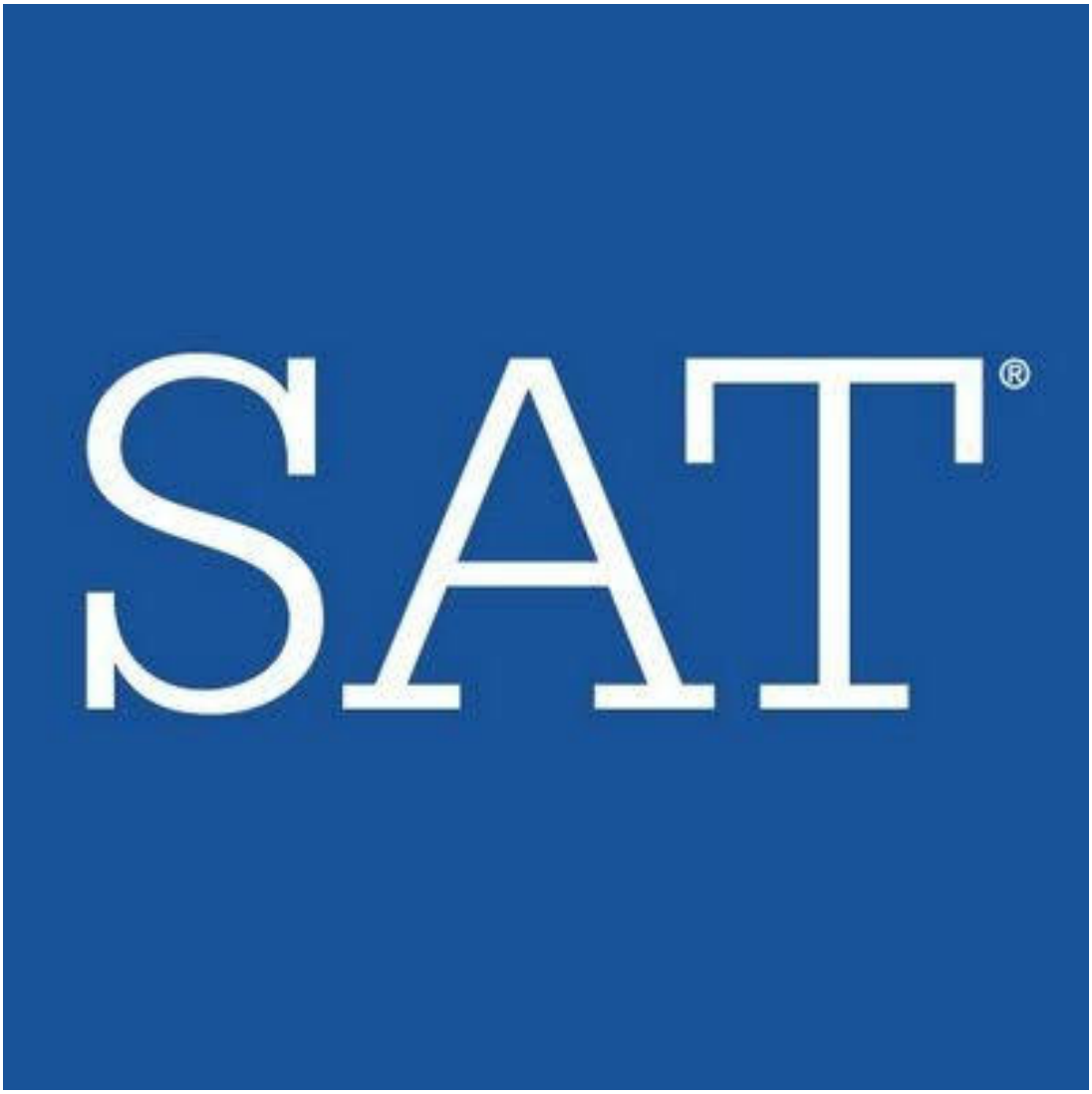
SAT Math

Key Points, Examples, and Practice Problems

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1. Systems of Linear Equations

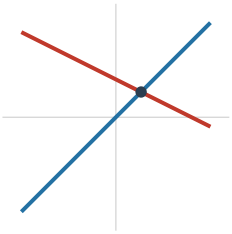
1.1. Key Points

For system of linear equations with 2 variables

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

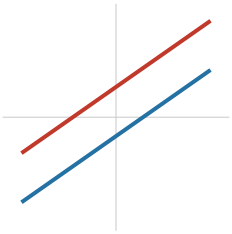
There will be 3 possibilities for its solution:



One solution

Lines intersect

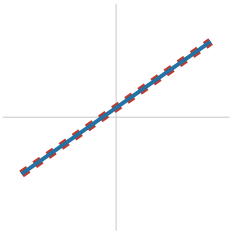
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$



No solution

Parallel lines

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$



Infinite solutions

Same line

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

1.2. Worked Examples

> 📝 **Example 1 Elimination**
>
> Solve:
>
> $$3x + 2y = 16$$
>
> $$5x - 2y = 0$$
>
> .

Solution.
Add equations (y cancels): $8x = 16 \Rightarrow x = 2$.

Back-sub: $2y = 16 - 6 = 10 \Rightarrow y = 5$. $(\mathbf{2},;\mathbf{5})$

> 📝 **Example 2 Infinite Solutions — Finding k**
>
> For what k does $2x - 4y = 8$ and $x - 2y = k$ have infinitely many solutions?

Solution.
For infinitely many solutions the two equations must represent the *same* line. Multiply the second equation by 2: $2x - 4y = 2k$. This must be identical to the first equation $2x - 4y = 8$, so we match the constant terms: $2k = 8 \Rightarrow \boldsymbol{k = 4}$.

📝 **Example 3 Substitution**

Solve: $y = -2x + 7$ and $3x + y = 13$.

Solution.
Substituting y to the second equation gives

$$3x + (-2x + 7) = 13 \Rightarrow x = 6, \Rightarrow y = -5$$

. So the solution is $(x, y) = (\mathbf{6}, \mathbf{-5})$.

1.3. Practice Problems

[Easy] **Problem 1**. If $y = -3x$ and $4x + y = 15$, what is x?

[Medium] **Problem 2**. $2a + 8b = 198$ and $2a + 4b = 98$. What is the value of b?

[Medium] **Problem 3**. For what value of c does $5x + 10y = 20$ and $x + 2y = c$ have **no** solution?

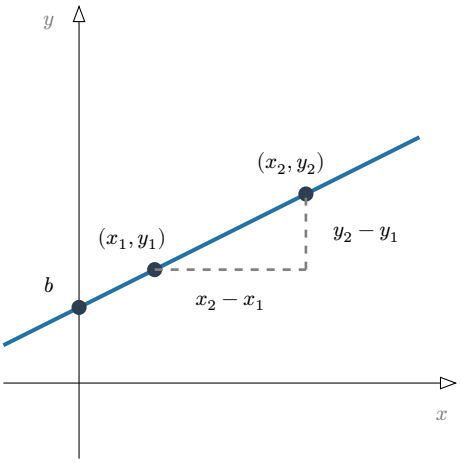
A) $c = 2$ B) $c = 4$ C) Any $c \neq 2$ D) Any $c \neq 4$

[Hard] **Problem 4**. $4x - 2y = 6$ and $-6x + 3y = k$ have infinitely many solutions. Find k.

[Hard] **Problem 5**. Mia buys x notebooks ($3 each) and y pens ($1.50 each). She spends exactly $18 and buys exactly 8 items. How many notebooks did she buy?

2. Graphs & Functions

2.1. Key Formulas



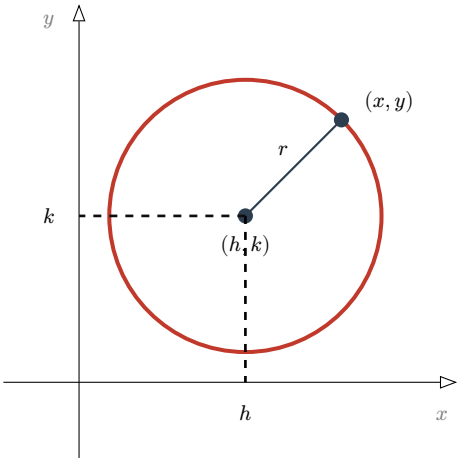
Line Properties

Slope: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

Equations: $y = mx + b \qquad y - y_1 = m(x - x_1)$

Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint: $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$



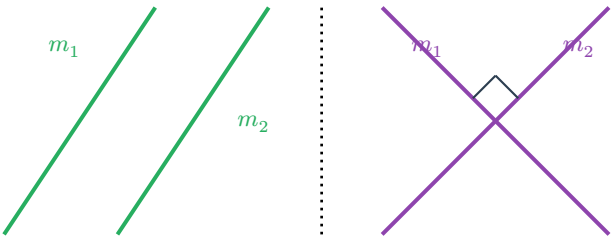
Circle Equation

You can think of a circle as a collection of points (locus) which has the same distance to the center (h, k)

Standard Form: $(x - h)^2 + (y - k)^2 = r^2$

Centre: (h, k)

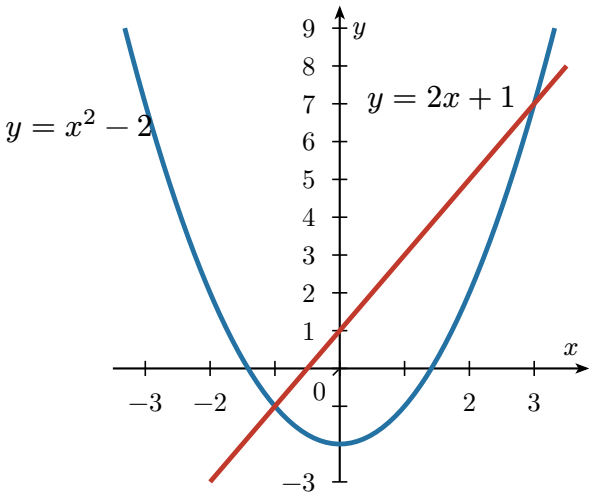
Radius: r



Line Relationships

Parallel Lines:
$m_1 = m_2$ (same slope, never intersect)

Perpendicular Lines:
$m_1 \cdot m_2 = -1$ (intersect at 90°)



Parabola Equations

Standard Form: $y = ax^2 + bx + c$
(y-int $= c$; axis $x = -\frac{b}{2a}$)

Vertex Form: $y = a(x-h)^2 + k$
(vertex (h, k); min/max $= k$)

Factored Form: $y = a(x - r_1)(x - r_2)$
(roots r_1, r_2 / x-intercepts)

Roots & Discriminant

$$x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$\Delta = b^2 - 4ac$

$\Delta > 0 \to$ 2 roots $\Delta = 0 \to$ 1 root (tangent)

$\Delta < 0 \to$ no real roots

Polynomial

Factor theorem: $p(r) = 0 \Leftrightarrow (x - r)$ is a factor.

Multiplicity 2 means the parabola **touches** (does not cross) x-axis at exactly one point.

In the diagram above, we have the line $2x + 1$ intercept the parabola at 2 points. An algebraic interpretation of this is that, if we substitute $y = 2x + 1$ to the parabola's quadratic equation, we will obtain 2 different roots/solutions ($\Delta > 0$).

Common terms:

- slope = gradien/kemiringan
- y-intercept, x-intercept = titik potong sumbu y, titik potong sumbu x
- tangent line = garis singgung
- vertex = titik puncak/lembah
- roots = akar/solusi polinomial

💡 Property 1 Factorisation Techniques

Difference of squares	$a^2 - b^2 = (a + b)(a - b)$
Perfect square (+)	$a^2 + 2ab + b^2 = (a + b)^2$
Perfect square (−)	$a^2 - 2ab + b^2 = (a - b)^2$
Quadratic trial	find p, q: $p + q = b$ and $pq = ac$ in $ax^2 + bx + c$
Grouping	split middle term $\to$ factor in pairs $\to$ common binomial

2.2. Examples

> 📝 **Example 4 Vertex & Minimum**
>
> Find the vertex and minimum value of $f(x) = 2x^2 - 8x + 3$.

Solution.

$$x_v = \frac{8}{4} = 2df(2) = 8 - 16 + 3 = -5$$

Vertex $(2, -5)$; minimum value $= \mathbf{-5}$.

> 📝 **Example 5 Factor Theorem — All Roots**
>
> $p(x) = x^3 + x^2 - 4x - 4$. Given $p(-2) = 0$, factor completely.

Solution.
Divide by $(x + 2)$: $p(x) = (x + 2)(x^2 - x - 2) = \boldsymbol{(x + 2)(x - 2)(x + 1)}$

Roots: $x = -2, 2, -1$.

> 📝 **Example 6 Perpendicular Line Through a Point**
>
> Line k: $y = 4x - 7$. Find the y-intercept of the perpendicular through $(8, 1)$.

Solution.
$m_{\text{perp}} = -1/4$. Point-slope: $y - 1 = -\dfrac{x - 8}{4} \Rightarrow y = -\dfrac{x}{4} + 3$.

Y-intercept $= \mathbf{3}$.

> 📝 **Example 7 Circle — Equation from Two Points**
>
> Centre $(-2, 3)$; passes through $(2, 6)$.

Solution.
$r = \sqrt{16 + 9} = 5 \quad \Rightarrow \quad \boldsymbol{(x + 2)^2 + (y - 3)^2 = 25}$

2.3. Practice Problems

[Easy] **Problem 6**. What is the slope of the line through $(-1, 5)$ and $(3, -3)$?

A) -2 B) -1 C) 1 D) 2

[Easy] **Problem 7**. The vertex of $y = -(x - 3)^2 + 7$ is at which point?

A) $(3, 7)$ B) $(-3, 7)$ C) $(3, -7)$ D) $(-3, -7)$

[Medium] **Problem 8**. The equation $x^2 + y^2 + 4x - 6y = 3$ represents a circle. What is its radius?

[Medium] **Problem 9**. For what value(s) of k does $x^2 + kx + 9 = 0$ have exactly one real solution?

A) $k = 3$ B) $k = 6$ C) $k = \pm 6$ D) $k = 9$

[Medium] **Problem 10**. At how many points do $y = x^2 - 3x - 4$ and $y = x - 4$ intersect?

A) 0 B) 1 C) 2 D) 3

[Hard] **Problem 11**. $p(x) = x^3 - 2x^2 - 5x + 6$ and $p(1) = 0$. Find all three roots.

[Hard] **Problem 12**. The parabola $y = x^2 - 6x + 5$ intersects the line $y = x - 1$ at two points. What is the positive difference between the x-coordinates of the two intersection points?

3. Optimization

3.1. Key Formulas

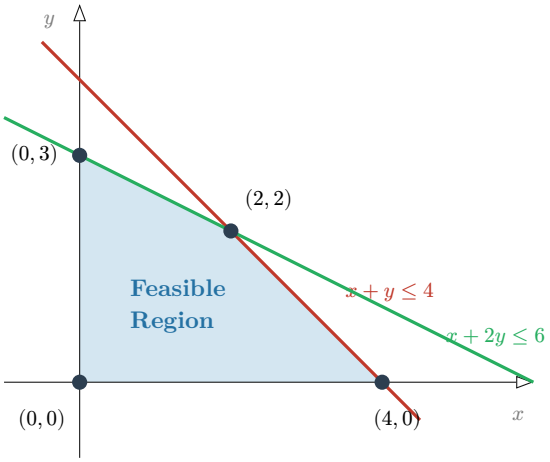
In an optimization, we mainly deal with these 3 things

- **Objective Function**
  The linear equation you want to maximize or minimize (e.g., $Z = 2x + 3y$).
- **Constraints**
  A system of linear inequalities that restrict the variables (e.g., time, budget, resources).
- **Feasible Region**
  The geometric area where **all** constraints overlap and are satisfied simultaneously.

For simple 2 variables linear programming problems, the following fact is true.

- **Corner-Point Principle**
  The optimal solution (maximum or minimum value) will **always** occur at one of the vertices (corners) of the feasible region.

3.2. Worked Examples



How to Optimize

Step 1: Graph the constraint inequalities.

Step 2: Shade the intersecting **Feasible Region**.

Step 3: Find coordinates of all **Corner Points**.

Step 4: Substitute corners into the objective function.

Step 5: Choose the highest/lowest result.

Example: Maximize $Z = 2x + 3y$ with constraint

$$x + y \leq 4$$

$$x + 2y \leq 6$$

$$x, y \geq 0$$

Corner Point	$Z = 2x + 3y$
$(0, 0)$	$2(0) + 3(0) = 0$
$(4, 0)$	$2(4) + 3(0) = 8$
$(0, 3)$	$2(0) + 3(3) = 9$
$(2, 2)$	$2(2) + 3(2) =$ **10** MAX

3.3. Practice Problems

[Easy] Problem 13. A bakery makes x loaves of bread and y cakes. Each loaf requires 2 hours of baking and each cake requires 3 hours. The total available baking time is at most 30 hours. Which inequality represents this constraint?

A) $2x + 3y \geq 30$ B) $2x + 3y \leq 30$ C) $3x + 2y \leq 30$ D) $x + y \leq 30$

[Medium] Problem 14. A factory produces two types of widgets, A and B. Each widget A yields a profit of \$4, and each widget B yields a profit of \$6. Labor constraints limit total production such that $x + 2y \leq 12$, where x is the number of widget A and y is widget B. What is the maximum possible profit $P = 4x + 6y$ if $x \geq 0$ and $y \geq 0$?

A) 24 B) 30 C) 36 D) 48

[Medium] Problem 15. A student sells handmade bracelets (x) and necklaces (y). Each bracelet takes 1 hour to make, and each necklace takes 2 hours. The student has at most 10 hours available per week. Additionally, they must make at least 3 necklaces. Which system of inequalities describes these restrictions?

A) $x + 2y \leq 10,\ y \geq 3$ B) $x + 2y \geq 10,\ y \leq 3$ C) $2x + y \leq 10,\ x \geq 3$ D) $x + 2y \leq 10,\ x \geq 3$

[Hard] Problem 16. A delivery company uses vans (x) and trucks (y). Vans can carry 20 packages and trucks can carry 50 packages. The company needs to deliver at least 200 packages total. Furthermore, due to driver availability, the total number of vehicles used cannot exceed 8. If the cost of operating a van is \$100 and a truck is \$300, how many of each vehicle should be used to **minimize** total operating cost while meeting all constraints?

A) 8 vans, 0 trucks B) 5 vans, 3 trucks C) 0 vans, 4 trucks D) 6 vans, 2 trucks

[Hard] Problem 17. A farmer has 100 acres of land to plant corn (x) and soybeans (y). Corn requires 2 hours of labor per acre and soybeans require 1 hour. Total available labor is 150 hours. If the profit per acre is \$300 for corn and \$200 for soybeans, what is the maximum profit the farmer can achieve?

A) \$20,000 B) \$22,500 C) \$25,000 D) \$30,000

4. Linear and Exponential Growth

4.1. Key Concepts

- **Linear growth** is modelled by the line equation

$$y = mx + b$$

  where slope m is its rate and b means initial value.

This is because each time step adds the same amount m, starting from b:

$$y_0 = b, \quad y_1 = b + m, \quad y_2 = b + 2m, \quad ..., \quad y_t = b + mt.$$

After t steps we have added m exactly t times, so $y = mt + b$.

- **Exponential growth** is modelled with constant **ratio** between consecutive terms.

Growing by a rate r means the new amount is the old amount plus r times the old amount:

$$y_{t+1} = y_t + ry_t = y_t(1 + r).$$

So each step multiplies by the same factor $(1 + r)$:

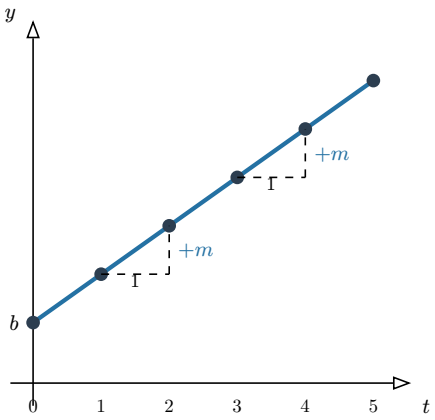
$$y_0 = a, \quad y_1 = a(1 + r), \quad y_2 = a(1 + r)^2, \quad ..., \quad y_t = a(1 + r)^t.$$

With the same idea,

- Decay is a negative rate, so the factor becomes $(1 - r)$.
- Doubling every T units has factor 2 per period. After t units there have been t/T periods, so $y = a \cdot 2^{t/T}$.
- Compound interest with n compounds per year uses rate r/n per period and nt periods in total, so $A = P(1 + r/n)^{nt}$.

Scenario	**Equation**	**Notes**
Growth	$y = a(1 + r)^t$	$r > 0$
Decay	$y = a(1 - r)^t$	$0 < r < 1$
Compound interest	$A = P(1 + r/n)^{nt}$	n = compounds/yr
Doubling time T	$y = a \cdot 2^{t/T}$	doubles every T units

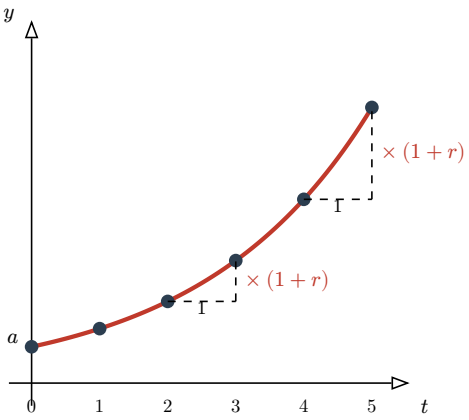
a = initial amount, r = rate (decimal), t = time.



Linear growth

$y = mx + b$

Same amount added each step



Exponential growth

$y = a(1 + r)^t$

Same factor multiplied each step

4.2. Examples

> 📝 **Example 8 Interpreting a Linear Model**
>
> A plumber charges $C = 75h + 50$ where $h =$ hours worked.

Solution.

- 75: rate of change — \$75 **per hour** of labour.
- 50: initial value — fixed **call-out fee** (charged even if $h = 0$).

> 📝 **Example 9 Exponential Growth — Population**
>
> A town of $8,000$ grows at 5%/year. Write $P(t)$ and find the population after 10 years.

Solution.

$$P(t) = 8000(1.05)^t$$

$$P(10) = 8000(1.05)^{10} \approx 8000 \times 1.629 \approx \mathbf{13,029}$$

4.3. Practice Problems

[Easy] **Problem 18**. In the model $C = 40d + 15$ (car rental: \$40/day + fee), what does 15 represent?

A) Daily rate B) Miles driven C) Fixed one-time fee D) Litres of fuel

[Easy] **Problem 19**. A bacteria culture of 200 doubles every 3 hours. Which expression gives the count after t hours?

A) $200 + 2t$ B) $200 \cdot 2^t$ C) $200 \cdot 2^{t/3}$ D) $200(1.02)^t$

[Medium] **Problem 20**. A town of $15,000$ grows at 4%/year. Write $P(x)$. In which year does it first exceed $20,000$? (Use $1.04^7 \approx 1.316$ and $1.04^8 \approx 1.369$.)

[Medium] **Problem 21**. Factor completely: $6x^2 + 11x - 10$.

[Hard] **Problem 22**. A car bought for \$24,000 depreciates at 12%/year. Find its value after 5 years. (Use $(0.88)^5 \approx 0.528$.)

[Hard] **Problem 23**. \$6,000 is invested at 8% annual interest compounded quarterly. Find the value after 3 years. (Use $(1.02)^{12} \approx 1.2682$.)

5. Data, Statistics, and Probability

5.1. Key Formulas

- **Mean**
  - $\overline{x} = \dfrac{\sum x_i}{n}$

  also: Sum $= n\overline{x}$
- **Median**
  - Middle value after sorting; average of two middles if n is even
  - The avearage of near middle values if n is odd
- **Range** $= x_{\max} - x_{\min}$
- **IQR** $= Q_3 - Q_1$
- **Probability**

$$P(E) = \frac{\text{favourable}}{\text{total}}$$

$$P(\overline{E}) = 1 - P(E)$$

One cool fact about mean and median is that mean is sensitive to outliers while median is not.

Here's how you read a frequency table

Score	**Freq. f**	**Rel. freq.**	**Cum. freq.**
60	3	15%	3
70	5	25%	8
80	7	35%	15
90	4	20%	19
100	1	5%	20
Total	**20**	**100%**	

From the given table above we can calculate that their mean and median by

$$\overline{x} = \frac{60(3) + 70(5) + 80(7) + 90(4) + 100(1)}{20} = \frac{1560}{20} = 78$$

For data with 20 values, we can average the 10th and 11th which both fall in the 80 group, so median = 80.

5.2. Worked Examples

📝 **Example 10 Mean from a Sum Constraint**

The mean of 6 numbers is 15. Five of them are $8, 12, 19, 22, 14$. Find the sixth.

Solution.
Sum $= 6 \times 15 = 90$. Known sum $= 75$. Sixth $= 90 - 75 =$ **15**.

📝 **Example 11 Weighted Mean**

10 girls (mean score 84) and 15 boys (mean score 76). Find the overall class mean.

Solution.

$$\overline{x} = \frac{10(84) + 15(76)}{25} = \frac{1980}{25} = \mathbf{79.2}$$

📝 **Example 12 Two-Way Table — Conditional Probability**

	Pass	**Fail**	**Total**
Studied	72	18	90
Did not	20	40	60
Total	92	58	150

What fraction of students who **studied** passed?

Solution.
$\frac{72}{90} = \frac{4}{5} = \mathbf{80\%}$

5.3. Practice Problems

[Easy] **Problem 24**. Ages: $24, 31, 45, 29, 31$. What is the median?

A) 29 B) 31 C) 32 D) 34

[Easy] **Problem 25**. The mean of $\{3, 7, x, 12, 18\}$ is 10. Find x.

[Medium] **Problem 26**. Using the frequency table above, what percentage of students scored **above** 75?

A) 35% B) 55% C) 60% D) 65%

[Medium] **Problem 27**. A bag has 5 red, 3 blue, and 2 green marbles. What is $P(\text{not red})$?

A) 1/5 B) 1/2 C) 2/5 D) 3/5

[Hard] **Problem 28**. 8 integers have a mean of 20. When one is removed, the mean of the remaining 7 is 18. What was removed?

[Hard] **Problem 29**. From the two-way table above, what percent of students who **failed** had not studied? (Nearest whole %.)

6. Geometry

6.1. Key Formulas

- **Angle sum**

$$\angle A + \angle B + \angle C = 180°$$

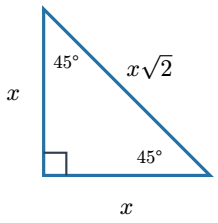
- **Pythagorean Theorem**

$$a^2 + b^2 = c^2$$

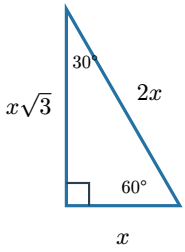
  (where c is the hypotenuse)

- **Special Right Triangles**
  - 30°-60°-90° → sides in ratio $1 : \sqrt{3} : 2$
  - 45°-45°-90° → sides in ratio $1 : 1 : \sqrt{2}$
- **Isosceles & Equilateral**
  - Isosceles → base angles equal; two legs equal
  - Equilateral → all angles 60°; all sides equal
  - All equilateral triangles are isosceles triangle, but the reverse is not true

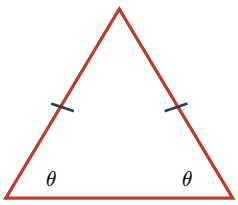
Common terms:
- Obtuse angle = sudut tumpul
- Acute angle = sudut lancip
- Right angle = sudut siku-siku



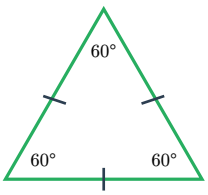
45°-45°-90°



30°-60°-90°



Isosceles

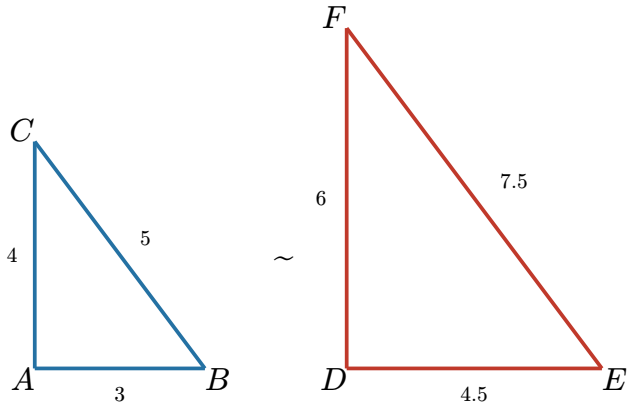


Equilateral

> 💡 **Property 2 Similar Triangles**
>
> We say $\triangle ABC$ is similar to $\triangle DEF$ denoted by $\triangle ABC \sim \triangle DEF$ when
>
> - two of their angles are the same (**AA**),
> - two of their sides are in proportion and the included angle is the same (**SAS**), or
> - all three pairs of corresponding sides are in proportion (**SSS**).
>
> Let k be the proportional factor, then

$$k = \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

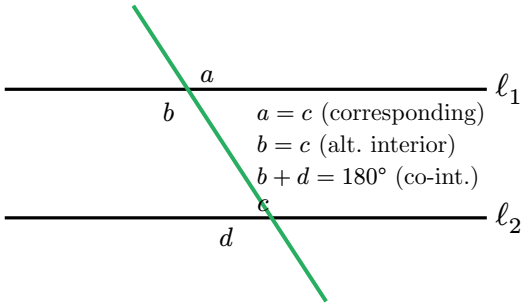
Corresponding angles are equal and the ratio of their area is k^2.



$\triangle ABC \sim \triangle DEF$ — scale factor $= 3:2$

💡 Property 3 Angle Chasing Parallel Lines

Vertical angles	equal
Supplementary	sum to 180°
Corresponding angles	equal when $\ell_1 \parallel \ell_2$ cut by a transversal
Alternate interior	equal when $\ell_1 \parallel \ell_2$ cut by a transversal
Co-interior (same-side)	sum to 180°



6.2. Worked Examples

📝 Example 13 Angle Sum in a Triangle

In $\triangle PQR$: $\angle P = 54°$ and $\angle Q = 2\angle R$. Find $\angle R$.

Solution.

$$54 + 2R + R = 180 \Rightarrow 3R = 126 \Rightarrow \angle \boldsymbol{R} = \mathbf{42°}$$

📝 **Example 14 Similar Triangles — Missing Side**

$\triangle ABC \sim \triangle DEF$; $AB = 6$, $DE = 9$, $BC = 8$. Find EF.

Solution.
Scale $= \dfrac{9}{6} = \dfrac{3}{2}$. $\quad EF = 8 \times \dfrac{3}{2} = \mathbf{12}$.

📝 **Example 15 Exterior Angle Theorem**

An exterior angle of a triangle is 118°. One non-adjacent interior angle is 63°. Find the other.

Solution.

$$118 = 63 + \theta \Rightarrow \boldsymbol{\theta = 55^\circ}$$

6.3. Practice Problems

[Easy] **Problem 30**. Two angles of a triangle are 37° and 94°. What is the third?

A) 49° B) 51° C) 57° D) 131°

[Easy] **Problem 31**. A 45°-45°-90° triangle has hypotenuse $8\sqrt{2}$. What is the length of one leg?

[Medium] **Problem 32**. $\triangle RST \sim \triangle XYZ$ with $RS = 4$, $XY = 10$, $ST = 6$. Find YZ.

[Medium] **Problem 33**. Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its alternate interior angle is $(5x - 9)^\circ$. Find x.

A) 8 B) 10 C) 12 D) 14

[Hard] **Problem 34**. Right $\triangle ABC$ has its right angle at C. Altitude $\overline{CD}$ meets $\overline{AB}$ at D with $AD = 4$ and $DB = 9$. Find CD.

A) $\sqrt{13}$ B) 5 C) 6 D) 6.5

[Hard] **Problem 35**. In $\triangle ABC$: $\angle A = 35^\circ$ and the exterior angle at vertex C is 100°. Find $\angle B$.

A) 45° B) 55° C) 65° D) 80°